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Investigation of interactions between phytase, protease, and anti-nutritional factors in soybeans

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ABSTRACT

The objective was to investigate interactions between phytase, protease, and anti-nutritional factors in soybeans. In Experiments 1 (in vitro), 2 (broiler chickens), and 3 (precision-fed roosters), trypsin inhibitor activity, growth performance, and amino acid (AA) digestibility were assessed. Diet 1 was a corn-soybean meal-based diet. Diet 2 contained 15% raw soybeans (RSB). Diets 3 and 4 were similar to diet 2, except 4,000 FTU/kg of phytase and 0.75% phytic acid were added, respectively. Experiment 3 (precision-fed roosters) contained 2 additional treatments (100% RSB $\pm$ 4,000 FTU/kg). In Experiments 4 and 5 (in vitro), trypsin inhibitor activity was measured in corn-soybean meal or corn-soybean meal-RSB diets $\pm$ protease. Experiment 6 was a broiler growth performance assay. Diet 1 was a corn-soybean meal-RSB (15%) diet. Diets 2 and 3 were the same as 1, except phytase (2,000 FTU/kg) or phytase + protease were added, respectively. Diet 4 was the same as diet 2, except phytase did not receive an energy matrix value. In Experiment 1 (in vitro), trypsin inhibitor activity was greater for the RSB + phytase diet compared with the RSB-no phytase diet ($P < 0.05$). In Experiment 2 (broilers), the addition of phytase to the RSB diet reduced BW gain and feed intake ($P < 0.05$). In Experiment 3 (precision-fed roosters), there was no effect of phytase on AA digestibility. In Experiments 4 and 5 (in vitro), protease reduced trypsin inhibitor activity in the corn-soybean meal diet ($P < 0.05$) but not the corn-soybean meal-RSB diet ($P > 0.05$). In Experiment 6 (broilers), phytase reduced BW gain and feed intake with and without an energy matrix value compared with the no phytase diet ($P < 0.05$). Protease did not effect growth performance ($P > 0.05$). In conclusion, phytase addition increased trypsin inhibitor activity and reduced growth performance when added to diets with RSB. Further, protease reduced trypsin inhibitor activity in vitro but did not ameliorate the effect of RSB + phytase in vivo.

Introduction

Soybean meal has an important role in poultry diets in the United States because it is high in protein (45 to 49% CP, as-fed), the amino acids (AA) are generally highly digestible (~86% or greater), and the balance of digestible AA resembles that of the bird's requirement for growth and protein accretion (Summers and Fisher, 1961; Kim et al., 2012; Aguirre et al., 2024). Although soybean meal contains a number of beneficial compounds or nutrients, soybeans contain a variety of anti-nutritional factors, such as trypsin inhibitors, lectins, oligosaccharides, and phytic acid, which can elicit negative effects on the growth and performance of birds (Liener, 1994; Ruiz et al., 2020). Of these anti-nutritional factors in soybeans, trypsin inhibitors receive the most

attention due to the reduction in protein digestibility that occurs if these factors are not adequately destroyed during the heating of soybean meal at the soybean crushing plant (Han et al., 1991; Douglas et al., 1999). Failure to reduce trypsin inhibitor levels below the tolerance threshold will lead to pancreatic hypertrophy, increased endogenous AA losses, reduced AA digestibility, and overall impairments in performance (Chernick et al., 1948; Green and Lyman, 1972; Clarke and Wiseman, 2005; Hoffman et al., 2019; Aderibigbe et al., 2021). Aside from using heat processing to inactivate trypsin inhibitors, another option that has been investigated to ameliorate reductions in digestibility from trypsin inhibitors is supplementation of proteases. There are 2 main mechanisms by which proteases may ameliorate the effects of trypsin inhibitors. The first is through compensation for the loss in endogenous

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protease activity through direct digestion of dietary proteins (Ghazi et al., 2002). The second is through direct digestion or breakdown of trypsin inhibitors themselves, thereby increasing the activity of endogenous proteases (Huo et al., 1993).

Aside from trypsin inhibitors, another anti-nutritional factor found in soybeans, as well as other plant-based ingredients, is phytic acid (myo-inositol 1,2,3,4,5,6-hexakis, **InsP₆**). Phytic acid or phytate, is a major source of P in poultry diets; however, little phytate-P is available to poultry unless exogenous phytase is supplemented in the diet (Nelson et al., 1971; Sebastian et al., 1996; Parsons and Rochell, 2024a). The ability of phytate to reduce the availability of nutrients, such as Ca, Zn, Cu, Fe, and AA has been well documented (Adeola, 1995; Adeola et al., 1995; Ravindran et al., 1999; Selle et al., 2009; Akter et al., 2018; Drysdale and Parsons, 2025). The inhibition of endogenous enzymes by phytate both in vitro and in vivo has also been reported (Singh and Krikorian, 1982; Liu et al., 2008, 2009). The study by Singh and Krikorian (1982) reported reduced trypsin activity of up to 46% when phytate was added in vitro. In the context of soybeans, the possibility exists that if phytate can bind to and inactivate trypsin, then it may also be able to bind to and inactivate trypsin inhibitors. Therefore, supplementation of phytase could potentially lead to impairments in digestibility and growth when added to diets with high levels of trypsin inhibitors, as the degradation of phytate by phytase could result in the release of active anti-nutritional factors in the gastrointestinal tract.

The objective of the present study was investigate interactions between phytase and anti-nutritional factors in soybeans and to determine if these interactions could be ameliorated by supplementation of protease.

Materials and methods

The protocols for this study were reviewed and approved by the Institutional Animal Care and Use Committee at the University of Arkansas (number 24013) and University of Illinois (number 20131).

Diets and design

Experiments 1 through 3 were conducted to investigate the potential of phytase to release active anti-nutritional factors from soybeans. The same diets were utilized across all three Experiments (Table 1). Diet 1 was a control corn-soybean meal-based mash diet that was formulated to be adequate in all nutrients. Diet 2 was the same as diet 1, except raw soybeans were added (15%). The level of 15% raw soybeans was selected to generate diets with high levels of anti-nutritional factors to increase the sensitivity of the bioassays for detecting potential release of active anti-nutritional factors by phytase. Diets 3 and 4 were the same as diet 2, except diet 3 contained 4,000 FTU/kg, while diet 4 contained 1.5% of an **InsP₆** solution (50% w/w in H₂O; Millipore Sigma, CAS: 83-86-3). The phytase used in diet 3 was Quantum Blue (*E. coli* derived 6-phytase, 10,000 FTU/kg; AB Vista, Marlborough, Wiltshire, UK). The use of 4,000 FTU/kg was chosen to ensure near complete degradation of phytate for this initial experiment to increase the sensitivity of the bioassay for detection of potential release of active anti-nutritional factors by phytase. The addition of 1.5% of the **InsP₆** solution in diet 4 was selected to achieve **InsP₆** levels similar to what would be present in diets with high phytate ingredients, such as wheat middlings. All diets were isocaloric and balanced for Ca, available P, and digestible amino acids; there was no adjustment made to account for potential reductions in nutrient digestibility due to anti-nutritional factors in the diets. In diet 3, phytase was given a matrix value for **AME_n** (83 kcal/kg), Ca (0.26%), available P (0.24%), and Na (0.06%) per AB Vista guidelines for 2,000 FTU/kg as these values would account for destruction of nearly all phytate in the diet. The analyzed AA levels in the diets are shown in Table 2.

In Experiment 1, the direct effect of exogenous phytase on trypsin inhibitor activity was evaluated using an in vitro assay. Dietary

Table 1
Composition of dietary treatments in Experiments 1, 2, and 3¹.

Ingredients, %	Dietary treatment			
	1	2	3	4
Corn	62.66	54.77	54.77	54.77
Soybean meal	31.50	22.50	22.50	22.50
Raw soybeans	–	15.00	15.00	15.00
Soybean oil	1.55	2.07	1.13	2.07
Limestone	0.91	0.90	0.96	0.90
Dicalcium phosphate	1.63	1.60	0.30	1.60
Phytase ²	–	–	0.04	–
Phytic acid solution ³	–	–	–	1.5
Solka Floc ⁴	–	1.50	3.85	–
DL-Met	0.28	0.28	0.28	0.28
L-Lys HCl	0.22	0.13	0.13	0.13
L-Thr	0.08	0.06	0.06	0.06
TiO ₂	0.50	0.50	0.50	0.50
Vitamin mix ⁵	0.05	0.05	0.05	0.05
Mineral mix ⁶	0.10	0.10	0.10	0.10
Choline chloride	0.10	0.10	0.10	0.10
Sodium bicarbonate	0.22	0.24	0.03	0.24
Sodium chloride	0.21	0.21	0.21	0.21
Analyzed and calculated values				
TUI / mg, analyzed	3.03	7.53	7.37	6.61
Phytase FTU/kg, analyzed	<50	78	3,780	<50
AME_n kcal/kg, calculated	3,025	3,025	3,025	3,025
CP %, analyzed	21.5	21.9	21.6	20.8
Ca %, calculated	0.84	0.84	0.84	0.84
NPP %, calculated	0.42	0.42	0.42	0.42
Na %, calculated	0.18	0.18	0.18	0.18
Dig. Lys %, calculated	1.12	1.20	1.12	1.12
Dig. Met + Cys %, calculated	0.85	0.85	0.85	0.85
Dig. Thr %, calculated	0.73	0.73	0.73	0.73
Dig. Arg %, calculated	1.20	1.23	1.23	1.23
Dig. Val %, calculated	0.85	0.85	0.85	0.85

¹Abbreviations: TUI = trypsin units inhibited; Dig. = digestible.
²Phytase was a 6-phytase, 10,000 FTU/kg (Quantum Blue, AB Vista, Marlborough, Wiltshire, UK).
³Phytic acid solution (myo-inositol 1,2,3,4,5,6-hexakis) was a 50/50 mixture with water.
⁴Powdered cellulose from International Corporation, Urbana, OH, U.S.
⁵Provided per kilogram of diet: vitamin A, 6,173 IU; vitamin D3, 4,409 ICU; vitamin E, 44 IU; vitamin B12, 0.01 mg; menadione, 1.20 mg; riboflavin, 5.29 mg; D-pantothenic acid, 7.94 mg; thiamine, 1.23 mg; niacin, 30.86 mg; pyridoxine, 2.20 mg; folic acid, 0.71 mg; biotin, 0.07 mg.
⁶Provided as milligrams per kilogram of diet: manganese, 100 mg; zinc, 100 mg; iron, 15 mg; selenium, 0.25 mg; copper, 15 mg; iodine, 1.2 mg.

Table 2
Analyzed amino acid concentrations in dietary treatments for Experiments 1,2, and 3 (%; as-fed).

Amino Acid	Dietary treatment			
	1	2	3	4
Asp	1.96	2.22	2.10	2.09
Thr	0.77	0.88	0.80	0.80
Ser	0.85	0.93	0.93	0.89
Glu	3.58	3.89	3.76	3.69
Pro	1.12	1.17	1.13	1.12
Gly	0.83	0.93	0.88	0.87
Ala	1.00	1.06	1.01	0.99
Cys	0.32	0.35	0.34	0.35
Val	0.95	1.05	0.99	0.99
Met	0.49	0.56	0.56	0.64
Ile	0.87	0.97	0.91	0.91
Leu	1.69	1.77	1.70	1.69
Tyr	0.69	0.75	0.71	0.72
Phe	0.98	1.07	1.01	1.01
Lys	1.28	1.40	1.28	1.28
His	0.54	0.59	0.56	0.56
Arg	1.29	1.48	1.37	1.37
Trp	0.22	0.21	0.22	0.20

treatments 2 and 3 from Table 1, diets containing 15% raw soybeans with and without 4,000 FTU/kg, were evaluated in this experiment. The *in vitro* assay followed a similar procedure to that described by Menezes-Blackburn et al. (2015) and contained 4 replications per treatment. Briefly, 4.6 g of feed was weighed into beakers with 27.4 ml of 50 mM sodium acetate-acetic acid buffer to yield a pH of 5. Samples were then placed in a heated water-shaker bath at 40°C for 30 min. After incubation, approximately 2,740 µl of 1 M HCl and 4,160 µl of 50 mM HCl-KCl buffer were added to lower the pH to 2. Beakers were then placed in a heated water-shaker bath at 40°C for 45 min. Lastly, approximately 5,200 µl of 1 M NaHCO₃ was added to raise the pH to 7 and breakers were placed in the heated water-shaker bath at 40°C for 60 min. Samples were then frozen, lyophilized, ground, and analyzed for trypsin inhibitor activity.

In Experiment 2, an *in vivo* assay with Cobb 500 male broiler chickens was used to investigate interactions between phytase and anti-nutritional factors. Broiler chickens were obtained from a commercial hatchery and were housed in battery cages with raised wire floors in an environmentally controlled room. The lighting schedule was as follows: 24 h of light on d0, 23L:1D from 1 to 4 d-of-age, 20L:4D from 5 to 15 d-of-age, and 18L:6D from d16 until the end of the experiment. Chickens were provided with a nutritionally complete corn-soybean meal-based crumbled starter diet until 7 d-of-age. Prior to the start of the experimental phase, birds were fasted overnight. On d 7, broilers were weighed and allotted to dietary treatments. A completely randomized design was used and birds were allotted to achieve a similar mean BW among treatments. There were 4 dietary treatments with 6 replicate pens of 5 broiler chickens per treatment that were provided to birds *ad libitum* (Table 1) from 7 to 21 d-of-age. These treatments consisted of: 1) control corn-soybean meal diet, 2) corn-soybean meal-raw soybean-based diet, 3) As 2 + 4,000 FTU/kg, and 4) As 2 + 0.75% InsP₆. On d 21, feed intake and BW were recorded per pen and birds were euthanized using CO₂ gas. The pancreas was collected from each bird to determine pancreatic weight. The ileal digesta were also collected from the distal half of the small intestine (Meckel's diverticulum to the ileocecal junction) via flushing. Digesta were stored frozen prior to being lyophilized and ground. Diets and freeze-dried ileal digesta were analyzed for Ti, AA, and InsP₆. Standardized ileal digestibility (SID) of AA was then calculated as described below:

SID of AA (%)

$$= [(AA \text{ diet} - AA \text{ ileal digesta} + \text{Endogenous AA}) / AA \text{ diet}] \times 100 \quad (1)$$

where AA diet = AA in the diet (%); AA ileal digesta = AA in the ileal digesta (%) × [Ti in the diet (%) / Ti in the ileal digesta (%)];

Endogenous AA = AA in the ileal digesta (%) × [Ti in the diet (%) / Ti in the ileal digesta (%)].

Endogenous AA were determined previously within our laboratory using broiler chickens fed a N-free diet.

The same equation [1] was used to calculate InsP₆ disappearance, except no correction was made for endogenous InsP₆ as this has previously been determined to be negligible in chickens (Parsons et al., 2023). In order to obtain sufficient sample for the InsP₆ analysis, digesta samples were pooled.

In Experiment 3, an *in vivo* precision-fed rooster assay (Sibbald, 1976) was used to evaluate interactions between phytase and anti-nutritional factors. Standardized AA digestibility was determined using cecectomized Single Comb White Leghorn roosters. Roosters were approximately 60 wk-of-age and weighed 2,200 g. Birds received 16 h of light and 8 h of dark. There were 6 dietary treatments evaluated in this experiment with 6 replicate pens of 1 individually caged rooster per treatment. Diets 1 through 4 were the same as those shown in Table 1: 1) control corn-soybean meal diet, 2) corn-soybean meal-raw soybean-based diet, 3) As 2 + 4,000 FTU/kg, and 4) As 2 + 0.75% InsP₆ (Table 1). Diets 5 and 6 were 100% raw soybean and 100% raw soybeans + 4000 FTU/kg, respectively. Roosters were fasted for 26 h prior to

being crop intubated with 25 g of feed. After roosters were precision-fed, they were placed in an individual cage with a tray placed underneath to quantitatively collect excreta for 48 h post-intubation. Excreta were then freeze-dried, weighed, and ground prior to being analyzed for AA. Standardized AA digestibility values were calculated as described by Engster et al. (1985) using AA excreted by fasted cecectomized roosters for the basal endogenous AA correction. The equation is shown below:

Standardized AA digestibility (%)

$$= [(AA \text{ intake} - AA \text{ excreted} + \text{Endogenous AA}) / AA \text{ intake}] \times 100 \quad (2)$$

where AA intake (g) = diet intake (g) × (AA in diet (%) / 100); AA excreted (g) = excreta output (g) × (AA in excreta (%) / 100). Endogenous AA = excreta output of fasted roosters (g) × (AA in excreta of fasted roosters (%) / 100)

Experiments 4 and 5 were conducted to assess the potential of a commercially available protease (Kerry, Tralee, Ireland) to reduce trypsin inhibitor activity in broiler chicken diets. Both experiments utilized the same *in vitro* digestion assay described previously for Experiment 1 and contained 4 replications per treatment. In Experiment

Table 3

Composition of dietary treatments in Experiment 6¹.

Ingredients, %	Dietary treatments			
	1	2	3	4 ²
Corn	55.9	55.9	55.9	55.9
Soybean meal	22.4	22.4	22.4	22.4
Raw soybeans	15.0	15.0	15.0	15.0
Dextrose	2.29	–	–	2.29
Soybean oil	0.71	0.71	0.71	0.71
Limestone	0.90	0.96	0.96	0.96
Dicalcium phosphate	1.60	0.30	0.30	0.30
Phytase ³	–	0.02	0.02	0.02
Protease ⁴	–	–	0.01	–
Solka floc ⁵	–	3.71	3.70	1.43
DL-Met	0.28	0.28	0.28	0.28
L-Lys HCl	0.13	0.13	0.13	0.13
L-Thr	0.06	0.06	0.06	0.06
Vitamin mix ⁶	0.05	0.05	0.05	0.05
Mineral mix ⁷	0.10	0.10	0.10	0.10
Choline chloride	0.10	0.10	0.10	0.10
Sodium bicarbonate	0.24	0.03	0.03	0.03
Sodium chloride	0.21	0.21	0.21	0.21
Calculated values				
TUI / mg, analyzed	7.6	8.1	7.4	7.8
Phytase FTU/kg, analyzed	<50	2,070	1,710	1,980
AME _n kcal/kg, calculated	3,025	3,025	3,025	3,025
CP %, analyzed	21.9	21.7	21.1	21.9
Ca %, calculated	0.84	0.84	0.84	0.84
NPP %, calculated	0.42	0.42	0.42	0.42
Na %, calculated	0.18	0.18	0.18	0.18
Dig. Lys %, calculated	1.12	1.12	1.12	1.12
Dig. Met + Cys %, calculated	0.85	0.85	0.85	0.85
Dig. Thr %, calculated	0.73	0.73	0.73	0.73
Dig. Arg %, calculated	1.23	1.23	1.23	1.23
Dig. Val %, calculated	0.85	0.85	0.85	0.85

¹Abbreviations: TUI = trypsin units inhibited; Dig. = digestible.

²Dietary treatment 4 was the same as 2 except phytase was given no matrix value for energy.

³Phytase was a 6-phytase, 10,000 FTU/kg (Optiphos Plus, Huvepharma Inc., Peachtree City, GA).

⁴Protease used obtained from Kerry (Tralee, Ireland). Protease recovery in diet 3 was 119% of the expected value.

⁵Powdered cellulose from International Corporation, Urbana, OH, U.S.

⁶Provided per kilogram of diet: vitamin A, 6,173 IU; vitamin D3, 4,409 ICU; vitamin E, 44 IU; vitamin B12, 0.01 mg; menadione, 1.20 mg; riboflavin, 5.29 mg; D-pantothenic acid, 7.94 mg; thiamine, 1.23 mg; niacin, 30.86 mg; pyridoxine, 2.20 mg; folic acid, 0.71 mg; biotin, 0.07 mg.

⁷Provided as milligrams per kilogram of diet: manganese, 100 mg; zinc, 100 mg; iron, 15 mg; selenium, 0.25 mg; copper, 15 mg; iodine, 1.2 mg.

4, there were 2 dietary treatments which consisted of a corn-soybean meal-based diet (same as diet 1 in Table 1) with and without 0.01% protease. In Experiment 5, there were 2 dietary treatments which consisted of a corn-soybean meal-raw soybean-based diet (same as diet 1 in Table 3) with and without 0.01% protease. Samples were frozen following the in vitro assay, lyophilized, ground, and analyzed for trypsin inhibitor activity.

Experiment 6 was conducted to confirm the results of Experiments 4 and to investigate the potential of protease to ameliorate the negative effects of anti-nutritional factors using an in vivo bioassay. Cobb 500 male broiler chickens were used. Broiler chickens were obtained from a commercial hatchery and were housed in batteries with raised wire floors in an environmentally controlled room. The lighting schedule was as follows: 24 h of light on d0, 23L:1D from 1 to 4 d-of-age, 20L:4D from 5 to 15 d-of-age, and 18L:6D from d16 until the end of the experiment. The experiment was conducted similarly to Experiment 3, where chickens were provided with a nutritionally complete corn-soybean meal-based crumbled starter diet until 7 d-of-age. Prior to the start of the experimental phase, birds were fasted overnight. On d 7, broilers were weighed and allotted to dietary treatments. A completely randomized design was used and birds were allotted to achieve a similar mean BW among treatments. There were 4 dietary treatments with 6 replicate pens of 5 broiler chickens per treatment that were provided to birds ad libitum from 7 to 21 d-of-age (Table 3). On d 21, feed intake and BW were recorded per pen and birds were euthanized using CO₂ gas. The pancreas was collected from each bird to determine pancreatic weight.

The dietary treatments in Experiment 6 were similar to Experiment 3, except that iso-caloricity was maintained across dietary treatments utilizing dextrose rather than soybean oil (Table 3). Diet 1 was a corn-soybean meal-raw soybean-based diet which contained 15% raw soybeans. Diets 2 was the same as diet 1, except 2,000 FTU/kg was added (Optiphos Plus, *E. coli* derived 6-phytase, 10,000 FTU/kg; Huvepharma Inc., Peachtree City, GA, U.S.) and phytase was given the same matrix values listed previously for Experiments 1 through 3. Diet 3 was the same as diet 2, except 0.1% protease was added (Kerry, Tralee, Ireland). Diet 4 was the same as diet 2, except phytase was not given a matrix value for AMEn. All diets were isocaloric and balanced for Ca, available P, and digestible amino acids.

Chemical analyses

Analyses were conducted at the University of Missouri-Columbia Experimental Station Chemical Laboratory (Columbia, MO, U.S.) to determine Ti using UV spectroscopy (Myers et al., 2004) and AA (Method 982.30 E [a, b, and c]; AOAC International, 2007). Analysis of trypsin units inhibited (TUI; AOCS method Ba12a-2020; Liu, 2021) and InsP₆ (Ellis et al., 1977) was performed by Eurofins (Des Moines, IA, U.S.). Phytase activity in feedstuffs was measured using a universal wet chemistry assay (ESC, Standard Analytical Method, SAM035-01 ISO FEFANA) by AB Vista (Plantation, FL, U.S.). Analysis of CP by measuring N via combustion (Method 990.03; AOAC International, 2007) was performed at the University of Arkansas Central Analytical Laboratory (Fayetteville, AR, U.S.).

Statistical analysis

SAS software (SAS Institute, Cary, NC) was used for all statistical analyses. Data from all experiments were analyzed using a 1-way ANOVA for a complete randomized design. Differences among means were evaluated using Fisher's least significant difference test (Carmer and Walker, 1985). For in vitro experiments (Experiments 1, 4, and 5), there were 4 replicates per treatment. In vivo experiments with broiler chickens (Experiments 2 and 6) contained 6 replicate pens of 5 broiler chickens per treatment, with the pen being the experimental unit. Experiment 3 (precision-fed rooster assay) contained 6 replicate pens of 1 individually caged rooster per treatment. The significance

value for all analyses was $P < 0.05$.

Results and discussion

Experiment 1, in vitro assay

The analyzed TUI following the in vitro assay from Experiment 1 is presented in Table 4. In this experiment, the trypsin inhibitor activity following the in vitro assay was greater ($P < 0.05$) for the high TUI diet + phytase compared with the high TUI diet without phytase. The relative TUI was then calculated by dividing the TUI from the diet that contained phytase by the diet that did not contain phytase. The relative trypsin inhibitor activity was increased ($P < 0.05$) by 8% from the addition of phytase. Singh and Krikorian (1982) reported that phytate was capable of reducing trypsin activity by up to 46% in vitro, demonstrating that phytate is capable of binding to and inactivating trypsin. It was hypothesized herein that if phytate can inactivate trypsin, there exists a possibility that phytate may also be capable of inactivating trypsin inhibitors. Should this be true, the destruction of phytate by exogenous phytase could subsequently result in the release of active trypsin inhibitors. The reduction in dietary Ca levels from inclusion of phytase could also contribute to higher trypsin inhibitor activity. In the small intestine, phytate can bind to proteins using Ca as an intermediate, thus reductions in dietary Ca levels caused by feeding phytase may also play a role in this response (Zhang et al., 2020). Further research is warranted to identify if the effects observed in Experiment 1 herein are affected by dietary Ca level or are Ca independent. Overall, results from Experiment 1 demonstrate in vitro that the incorporation of exogenous phytase into the diet can result in increased trypsin inhibitor activity.

Experiment 2, broiler chicken assay

In vivo growth performance, pancreatic weights, and InsP₆ disappearance from Experiment 2 are presented in Table 5. The InsP₆ disappearance ranged from 1 to 14% for dietary treatments 1, 2, and 4 which contained no exogenous phytase. Dietary treatment 3, which contained exogenous phytase (4,000 FTU/kg), exhibited 93% InsP₆ disappearance. This, combined with the analyzed level of 3,780 FTU/kg in the diet (Table 1), confirmed that the exogenous phytase in this experiment was active and was effective at degrading the phytate in the diet. The BW gain and gain:feed was the highest ($P < 0.05$) for the control treatment with no raw soybeans (diet 1; Table 5). The addition of InsP₆ to diets with high levels of trypsin inhibitors (diets 2 vs. 4) did not result in significant changes to BW gain, feed intake, or gain:feed ($P > 0.05$). The addition of phytase (diet 3), however, resulted in reductions in BW gain ($P < 0.05$), where BW gain was reduced by a mean of 62 g/chick when compared to the high TUI diet without phytase (diet 2). Furthermore, the addition of phytase (diet 3) reduced feed intake ($P < 0.05$) but not gain:feed ($P > 0.05$) when compared with diet 2.

It cannot be determined from this experiment whether the reduction in BW gain preceded the reduction in feed intake or vice versa; however, additional experiments conducted within our laboratory indicate that the reduction in BW appears to precede the reduction in feed intake

Table 4
Analyzed trypsin inhibitor activity after in vitro digestion in Experiment 1^{1,2}.

Dietary treatment ³	TUI/mg	Relative TUI (%)
2. High TUI	7.25	100
3. High TUI + phytase	7.86	108
SEM	0.100	–
P-value	0.005	–

¹Abbreviations: TUI = Trypsin units inhibited, measured using AOCS method (Liu et al., 2021).

²Values are means of 4 replicates.

³Dietary treatments from Table 1.

Table 5Growth performance of broiler chickens in Experiment 2¹.

Dietary treatment	BW gain (g/chicken)	Feed intake (g/chicken)	Gain:feed (g/kg)	Pancreas (g/chicken)	Pancreas (% of BW)	InsP ₆ D ² (%)
1. Control	720 ^a	999 ^a	721 ^a	3.11 ^c	0.37 ^b	0.85
2. High TUI	646 ^b	978 ^{ab}	661 ^b	5.07 ^a	0.66 ^a	4.61
3. As 2 + 4,000 FTU/kg	584 ^c	892 ^c	655 ^b	4.53 ^b	0.64 ^a	93.00
4. As 2 + InsP ₆	621 ^{bc}	924 ^{bc}	672 ^b	4.99 ^a	0.67 ^a	13.81
SEM	16.9	20.4	9.1	0.121	0.011	–
P-value	<0.001	0.007	<0.001	<0.001	<0.001	–

^{a-c}Means within a column with no common superscript differ ($P < 0.05$). Values are means of 6 pens of 5 chickens from 7 to 21 d-of-age.¹Abbreviations: TUI = trypsin units inhibited; InsP₆ = myo-inositol 1,2,3,4,5,6-hexakis; InsP₆D = InsP₆ disappearance.²Digesta were pooled to obtain sufficient sample for the InsP₆ analysis.

when phytase is added to diets with 15% raw soybeans (R. Drysdale, unpublished data). Pacheco et al. (2014) also observed that a reduction in BW proceeded the reduction in feed intake when the trypsin inhibitor content of the diet increased. The cause for the reduced BW gain herein (Experiment 2; Table 5) could be the result of multiple factors. First, phytase was given a matrix value for energy per manufacturer recommendations (83 kcal/kg). Previous studies have demonstrated increased metabolizable energy and carcass energy retention from supplementation of exogenous phytase (Cowieson et al., 2006; Pirgozliev et al., 2011). If phytase was not truly releasing the amount of energy it received credit for in the formulation matrix, however, this would account for some of the BW gain reduction observed when phytase was added to the diet. Another possible cause for the reduced BW gain when phytase was added may be due to release of active anti-nutritional factors. As discussed previously for Experiment 1 (Table 4), trypsin inhibitor activity was observed to be greater for diet 3 (phytase) compared with the no phytase diet (diet 2) following an in vitro digestion. Thus, the addition of phytase may have led to increased trypsin inhibitor activity in the gastrointestinal tract. Based on these observed effects of exogenous phytase in Experiment 2, it could be expected that adding InsP₆ to the diet may improve growth performance by reducing trypsin inhibitor activity, however no effect was observed on growth performance (Table 5). This could be due to the added InsP₆ binding other nutrients or enzymes in addition to anti-nutritional factors. It should also be noted that the added InsP₆ was a concentrated acid, which may be a confounding factor as this could have altered gastrointestinal pH.

The pancreatic weights both as g/chicken and expressed as a % of BW are presented in Table 5. The control treatment (diet 1) had the lowest pancreatic weight ($P < 0.05$; Table 5). This was expected as the remaining dietary treatments contained 15% raw soybeans. An experiment conducted by Perilla et al. (1997) showed similar results to the present study, where there was depression in BW gain and an increase in pancreas weight caused by feeding raw soybeans. There were no differences in pancreas weight expressed as both g/chicken and % of BW for the high TUI diet with and without added InsP₆ (diets 2 vs. 4; $P > 0.05$). Dietary treatment 3, which contained exogenous phytase, resulted in reduced ($P < 0.05$) pancreatic weights compared with the high TUI diet with no phytase (diet 2) when expressed as g/chicken. There was no difference between these 2 treatments ($P > 0.05$), however, when the weights were expressed as a % of BW; therefore, the lower pancreatic weights expressed as g/chicken were likely caused by the smaller size of the birds and not the dietary treatments per se.

The SID of AA for Experiment 2 are presented in Table 6. The SID of AA were significantly higher ($P < 0.05$) for the control treatment (diet 1) compared with diets with 15% raw soybeans (diets 2 through 4) for a number of amino acids, such as Lys, Cys, Val, and Trp. The overall mean SID of AA was also greater for diet 1 compared with diets with raw soybeans, where the mean SID of AA for the control diet (diet 1) was 85.5% compared with 80.0% for diets 2 through 4. When comparing diet 2 (high TUI diet – no phytase) with diet 3 (High TUI diet with phytase) and diet 4 (high TUI diet with added InsP₆), no differences were observed ($P > 0.05$). The cause for the reduced BW gain from the addition of exogenous phytase but lack of effect on SID of AA may be due

Table 6

Standardized ileal digestibility of amino acids in broiler chickens in Experiment 2 (%).

Amino acid	Dietary treatment ¹				SEM	P-value
	1	2	3	4		
Asp	85.2 ^a	80.4 ^b	81.4 ^b	80.1 ^b	1.10	0.025
Thr	78.9	74.4	73.9	72.6	1.55	0.069
Ser	85.7 ^a	79.1 ^b	80.4 ^b	78.1 ^b	1.42	0.010
Glu	89.4	86.6	87.3	86.2	0.88	0.112
Pro	84.8 ^a	80.5 ^b	80.5 ^b	80.3 ^b	1.14	0.048
Gly	80.9	75.8	76.0	75.6	1.38	0.057
Ala	85.6 ^a	80.7 ^b	80.6 ^b	80.0 ^b	1.21	0.022
Cys	73.8 ^a	64.1 ^b	64.0 ^b	66.0 ^b	2.21	0.026
Val	86.1 ^a	79.9 ^b	80.0 ^b	79.9 ^b	1.28	0.009
Met	93.2	90.3	90.7	91.8	0.74	0.072
Ile	86.2 ^a	79.6 ^b	79.9 ^b	79.2 ^b	1.25	0.004
Leu	86.8 ^a	80.9 ^b	81.2 ^b	80.6 ^b	1.19	0.008
Tyr	86.5 ^a	79.5 ^b	80.1 ^b	79.2 ^b	1.25	0.003
Phe	87.2 ^a	82.3 ^b	82.8 ^b	81.4 ^b	1.17	0.019
Lys	88.2 ^a	84.6 ^b	85.2 ^b	83.4 ^b	0.96	0.022
His	87.2 ^a	83.4 ^b	82.7 ^b	83.3 ^b	1.04	0.038
Arg	89.6	86.7	87.4	86.1	0.84	0.060
Trp	83.0 ^a	70.3 ^b	73.3 ^b	70.8 ^b	1.67	<0.001
Mean	85.5 ^a	79.9 ^b	80.4 ^b	79.7 ^b	1.20	0.015

^{a-b}Means within a column with no common superscript differ ($P < 0.05$). Values are means of 6 pens of 5 chickens at 21 d-of-age.¹Dietary treatments from Table 1: 1) control corn-soybean meal; 2) High trypsin inhibitor; 3) As 2 + 4,000 FTU/kg; 4) As 2 + myo-inositol 1,2,3,4,5,6-hexakis.

to the fact that ileal digesta were collected on the final day of the experiment, 14 days after birds were placed on experimental diets. As seen in Table 5, chickens fed diets 2 through 4 that contained 15% raw soybeans had substantial increases in the weight of the pancreas; therefore, the adaptation by the birds to increase the size of the pancreas to overcome the trypsin inhibitors in the diet may have masked effects of the exogenous phytase and InsP₆ on AA digestibility that were present early in the experimental phase. Erdaw et al. (2017) found similar effects in chickens, where it was reported that increasing the raw soybean content of diets from 0 to 20% resulted in the greatest effect on the weight of the pancreas expressed as a % of BW at 10 d-of-age compared with 24 and 35 d-of-age. Furthermore, a mean difference of only approximately 2 percentage units was observed for AA digestibility values in that study for the control diet compared with the diet with 10% raw soybeans at 24 d-of-age. Collections from earlier time points are thus warranted for future studies where interactions between exogenous phytase, anti-nutritional factors, and SID of AA are assessed as this may better align with and explain the results observed for growth performance.

Experiment 3, precision-fed rooster assay

Standardized AA digestibility values determined in precision-fed roosters in Experiment 3 are presented in Table 7. Dietary treatments 1 through 4 contained higher AA digestibility values ($P < 0.05$) compared to treatment 5 (100% raw soybeans) and treatment 6 (100%

Table 7
Standardized amino acid digestibility in precision-fed roosters in Experiment 3¹ (%).

Amino acid	Dietary treatment						SEM	P-value
	1	2	3	4	5	6		
Asp	88.9 ^a	86.0 ^a	85.7 ^a	86.4 ^a	63.3 ^b	68.1 ^b	2.21	0.001
Thr	87.8 ^a	85.6 ^a	84.3 ^a	85.1 ^a	56.8 ^b	61.2 ^b	2.60	0.001
Ser	89.3 ^a	86.7 ^a	86.6 ^a	86.0 ^a	58.3 ^b	63.5 ^b	2.80	0.001
Glu	92.0 ^a	89.7 ^a	89.8 ^a	90.0 ^a	69.0 ^b	74.0 ^b	2.13	0.001
Pro	91.0 ^a	88.8 ^a	89.0 ^a	89.1 ^a	64.3 ^b	68.7 ^b	2.18	0.001
Ala	86.5 ^a	84.0 ^a	83.3 ^a	84.7 ^a	56.6 ^b	61.5 ^b	2.24	0.001
Cys	85.3 ^a	83.8 ^a	82.5 ^a	84.7 ^a	59.3 ^b	60.9 ^b	2.59	0.001
Val	89.0 ^a	84.5 ^a	84.3 ^a	85.6 ^a	56.9 ^b	60.5 ^b	2.25	0.001
Met	93.4 ^a	92.2 ^a	92.2 ^a	94.0 ^a	67.5 ^b	69.8 ^b	1.39	0.001
Ile	88.5 ^a	84.1 ^a	83.7 ^a	85.2 ^a	58.0 ^b	62.5 ^b	2.25	0.001
Leu	90.7 ^a	86.8 ^a	86.7 ^a	87.9 ^a	58.5 ^b	63.5 ^b	2.19	0.001
Tyr	87.5 ^a	85.1 ^a	84.0 ^a	85.7 ^a	61.2 ^b	65.9 ^b	2.20	0.001
Phe	89.7 ^a	86.5 ^a	86.4 ^a	87.6 ^a	58.8 ^b	64.3 ^b	2.39	0.001
Lys	87.8 ^a	86.4 ^a	85.0 ^a	85.9 ^a	65.4 ^b	70.1 ^b	1.86	0.001
His	85.5 ^a	84.7 ^a	83.9 ^a	86.0 ^a	63.1 ^b	66.5 ^b	1.96	0.001
Arg	93.5 ^a	91.1 ^a	90.9 ^a	91.4 ^a	70.0 ^b	74.2 ^b	1.98	0.001
Trp	93.5 ^a	88.1 ^b	89.3 ^{ab}	89.1 ^{ab}	69.3 ^c	71.9 ^c	1.48	0.001
Mean	89.5 ^a	85.1 ^a	84.5 ^a	86.3 ^a	61.2 ^b	61.7 ^b	2.08	0.001

^{a-c}Means within a column with no common superscript differ ($P < 0.05$). Values are means of 6 pens of 1 individually caged cecetomized rooster.
¹Dietary treatments from Table 1: 1) control corn-soybean meal; 2) High trypsin inhibitor; 3) As 2 + 4,000 FTU/kg; 4) As 2 + myo-inositol 1,2,3,4,5,6-hexakis. Dietary treatments 5 and 6 were 100% raw soybeans and 100% raw soybeans + 4,000 FTU/kg, respectively.

raw soybeans + phytase). Although dietary treatment 1 (control corn-soybean meal diet) was not significantly different from diets 2 through 4 which contained 15% raw soybeans ($P > 0.05$), which was confirmed by conducting contrasts of diet 1 vs. 2 through 4 (data not shown), diet 1 did contain numerically higher AA digestibility values for all AA. For example, mean AA digestibility for diet 1 was 89.5%, whereas diets 2 through 4 ranged from 84.5 to 86.3%. Overall, there was no effect of phytase or InsP₆ on standardized AA digestibility in precision-fed roosters. The lack of effect of phytase and InsP₆ on AA digestibility in for dietary treatments 2 through 4 in precision-fed roosters is in good agreement with broiler chickens in Experiment 2. Although the short bioassay length in Experiment 3 would prevent pancreatic adaptations by the birds to the trypsin inhibitors, the fully mature pancreas of adult roosters is likely more capable of overcoming the single exposure to diets with high levels of trypsin inhibitors compared with ad libitum-fed broiler chickens. It should also be noted that much remains to be elucidated regarding the sensitivity of the precision-fed rooster assay for evaluation of the effects of exogenous enzymes on nutrient digestibility as few studies have been conducted with this bioassay (Assadi et al., 2011; Anderson et al., 2022); albeit, it has been shown that this bioassay is sensitive for assessing the effects of exogenous phytase on phytate degradation (Parsons et al., 2023; Parsons and Rochell, 2024b).

Experiments 4 and 5, in vitro assay

Two additional in vitro assays were conducted to determine if a commercially available protease could degrade trypsin inhibitors (Table 8). When protease was added to a control corn-soybean meal-based diet with no raw soybeans in Experiment 4, there was a reduction ($P < 0.05$) in trypsin inhibitor activity. The relative reduction in trypsin inhibitor activity was 11%. In Experiment 5, protease was added to a high TUI diet that contained 15% raw soybeans. In contrast to Experiment 4, when protease was added to a diet with high levels of trypsin inhibitors, there was no change in trypsin inhibitor activity ($P > 0.05$), being 8.14 TUI/mg without protease and 8.12 TUI/mg with protease. In vivo, proteases may improve growth performance through a variety of mechanisms, such as breaking down undigested proteins, compensating for loss in endogenous protease activity caused by factors such as trypsin inhibitors, reducing endogenous amino acid losses, and directly degrading specific anti-nutritional factors (Cowieson and Roos, 2016; Aderibigbe et al., 2020; Wedekind et al., 2020; Liu et al., 2024). The protease evaluated herein appears to possess some capacity to directly

Table 8
Analyzed trypsin inhibitor activity after in vitro digestion in Experiments 4 and 5.

Dietary treatment	Digesta TUI/mg ²	Relative TUI (%)
Experiment 4 ³		
Control corn-SBM	1.89	100
Control corn-SBM + protease	1.68	89
SEM	0.037	–
P-value	0.029	–
Experiment 5 ⁴		
High TUI	8.14	100
High TUI + protease	8.12	100
SEM	0.098	–
P-value	0.900	–

¹Abbreviations: SBM = soybean meal; TUI = Trypsin units inhibited, measured using AOCS method (Liu et al.,2021).
²Values are means of 4 replicates.
³Dietary treatments were the same as diet 1 from Table 1 with and without 0.01% protease (Kerry, Tralee, Ireland).
⁴Dietary treatments were the same as diet 1 from Table 3 with and without 0.01% protease (Kerry, Tralee, Ireland).

degrade trypsin inhibitors as was observed in Experiment 4; however, the lack of response in diet with high levels of trypsin inhibitors suggests that this capability may be dependent on the enzyme dose and anti-nutritional factor level. In fact, the protease evaluated herein was a serine-protease and thus it is possible that the high levels of trypsin inhibitors in Experiment 4 could have led to the direct inhibition of the exogenous protease.

Experiment 6, broiler chicken assay

Growth performance from Experiment 6 is presented in Table 9. The feed intake, BW gain, gain:feed, and pancreatic weights were the highest diet 1, which was the high TUI diet with no phytase and no protease. Similar to Experiment 2, the additional of phytase (diet 2; 2,000 FTU/kg) reduced ($P < 0.05$) BW gain, feed intake, and gain:feed. When phytase was added but given no matrix value for energy (diet 4), BW gain and feed intake were greater ($P < 0.05$) compared with diet 2, but BW gain was still lower compared with diet 1 (no phytase; $P < 0.05$). The pancreas expressed as both g/chicken and as a % of BW was greater for

Table 9Growth performance of broiler chickens in Experiment 6¹.

Dietary treatment	BW gain (g/chicken)	Feed intake (g/chicken)	Gain: feed (g/kg)	Pancreas (g/chicken)	Pancreas (% of BW)
1. High TUI	838 ^a	1259 ^a	666 ^a	6.27 ^a	0.65 ^a
2. As 1 + 2000 FTU/kg	716 ^c	1120 ^c	640 ^b	4.90 ^b	0.59 ^b
3. As 2 + protease	712 ^c	1105 ^c	643 ^b	4.87 ^b	0.55 ^b
4. As 2 but no energy matrix for phytase	786 ^b	1195 ^b	658 ^{ab}	5.29 ^b	0.57 ^b
SEM	15.7	21.5	6.0	0.166	0.017
P-value	<0.001	<0.001	0.025	<0.001	0.007

^{a-c}Means within a column with no common superscript differ ($P < 0.05$). Values are means of 6 pens of 5 chickens from 7 to 21 d-of-age.

¹Abbreviations: TUI = trypsin units inhibited.

diet 1 compared with the other dietary treatments ($P < 0.05$). Although expressing the weight of the pancreas as a % of BW helps to control the confounding effect of weight differences among treatment, the large differences in BW of chickens on diet 1 compared with diets 2 through 4 may still be a confounding effect pancreatic weight comparisons. The greater BW gain for diet 4 (phytase – no energy matrix given to the enzyme) compared with diet 2 (phytase with an energy value) was expected as diet 4 should contain a greater ME value compared with diet 2 assuming phytase was releasing energy in the diet. Interestingly, even when phytase was given no matrix value for energy (diet 4), BW gain and feed intake were still reduced ($P < 0.05$) compared with the high TUI diet without phytase and protease (diet 1). This demonstrates that although the energy matrix value assigned to the phytase could be partially responsible for the growth performance effects if the assumed value is overestimated, another interaction must also be occurring to yield the growth depressions observed in Experiment 6.

There was no effect of protease ($P > 0.05$) on BW gain, feed intake, gain:feed, and pancreatic weight (Table 9). When protease was added to diets that contained high levels of trypsin inhibitors (15% raw soybeans) in vitro in Experiment 5, there was no change in trypsin inhibitor activity (Table 8). This was in agreement with Experiment 6 in broiler chickens where no effects of protease were observed in vivo on growth performance and pancreatic weight ($P > 0.05$; Table 8). The effectiveness of protease in poultry diets can be variable and may be dependent of the type of protease, the feed ingredients used in formulation, interactions with other enzymes, and protease dosage (Cowieson and Roos, 2016; Lee et al., 2018). The dose of the protease may be of particular importance in Experiment 6 (Table 9), as effects on growth performance may have been observed if higher levels of protease were added. As observed in vitro in Experiments 5 and 6 (Table 8), the protease evaluated in the present study was able to reduce trypsin inhibitor activity in diets with lower levels of inhibitors, but was relatively ineffective in diets with high levels of inhibitors (15% raw soybeans). More research is warranted evaluating more doses of exogenous proteases in diets with differing levels of anti-nutritional factors. Another consideration when assessing direct degradation of trypsin inhibitors by protease is the type of inhibitor that is present. The Bowman-Birk inhibitor and Kunitz inhibitor differ in structure, and thus may be more effectively degraded by different proteases (Papadoitsis and Wilson, 1991). Assessment of degradation of Kunitz vs. Bowman-Birk inhibitors by exogenous protease was not evaluated herein, but may be an interesting comparison for future studies.

Results from the experiments herein are important for commercial poultry nutritionists. The increased anti-nutritional factor activity, particularly trypsin inhibitors, that resulted from feeding high levels of phytase places increased emphasis on soybean meal quality. Feeding high doses of phytase in diets with soybean meals that are already

reaching the upper tolerance threshold for trypsin inhibitors could result in growth performance issues. Exogenous proteases may then offer an option by which to counteract these effects (Lee et al., 2018; Liu et al., 2024). Another interesting consideration is if the increase in trypsin inhibitor activity from feeding high doses of phytase could potentially be part of the extra-phosphoric effects of phytase observed in practical diets with lower levels of anti-nutritional factors than what was evaluated herein (Gehring et al., 2013; Walk et al., 2013). There may exist a window in which increases in trypsin inhibitor activity in diets with low levels of trypsin inhibitors may stimulate more rapid pancreatic growth during the first week of life, leading to improvements in nutrient digestibility and growth performance. More work is needed to investigate the interactions reported herein in commercially relevant diets.

In conclusion, results from this study indicate that utilization of high doses of phytase can increase trypsin inhibitor activity in poultry diets containing high levels of raw soybeans. Trypsin inhibitor activity was increased ($P < 0.05$) by the addition of phytase to diets with 15% raw soybeans in vitro. Furthermore, BW gain and feed intake were reduced ($P < 0.05$) in vivo from inclusion of phytase. The addition of protease to a lower TUI diet in vivo reduced trypsin inhibitor activity ($P < 0.05$), but no effects were observed in vitro in higher TUI diets with 15% raw soybeans or in vivo in diets with 15% raw soybeans ($P > 0.05$). This study provides new insights into interactions between anti-nutritional factors, phytase, and protease. The trypsin inhibitor-phytase interactions identified herein are particularly important in regard to soybean meal quality, as the use of high doses of phytase may shift the tolerance threshold for trypsin inhibitors in soybean meal for poultry. Further research is warranted to investigate these interactions in diets with lower levels of anti-nutritional factors with additional time points to better understand the effects of these compounds on nutrient digestibility, pancreatic growth, and growth performance of broiler chickens.

Author contribution

R. L. Drysdale is credited with conducting the experiment, data analysis, and writing the original draft of the manuscript. S. W. Hufford was responsible for overseeing the care of the animals, diet mixing, and experimental design. G. L. Engelke and B. W. Parsons are credited with conceptualization of the study, experimental design, and review & editing of the manuscript.

Disclosures

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Benjamin W. Parsons reports financial support was provided by United Soybean Board. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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